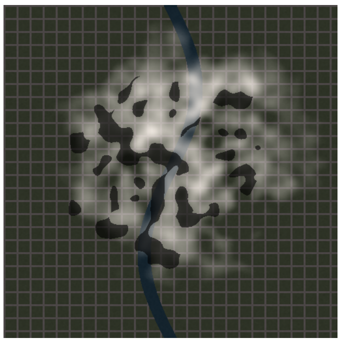
■■ CloudClear AI — Remote Sensing Quality Inspection Report

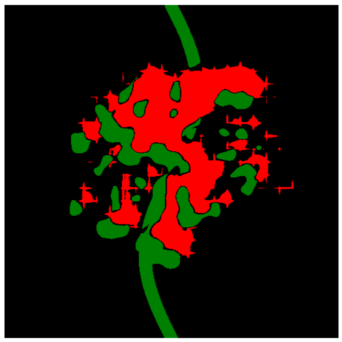
Multi-Modal Cloud Removal & Spectral Reconstruction Product Quality Report • ISRO Remote Sensing Standard

Scene ID:	scene_02_maharashtra_pune	**Region / AOI:**	West Bengal
Acquisition Date:	2024-05-12	**Sensors:**	Sentinel-2 + Sentinel-1 SAR
CRS:	EPSG:4326	**Resolution:**	0.0002929687500000111 m
Selected Strategy:	Adaptive (C1)	**Overall Rating:**	**Good**

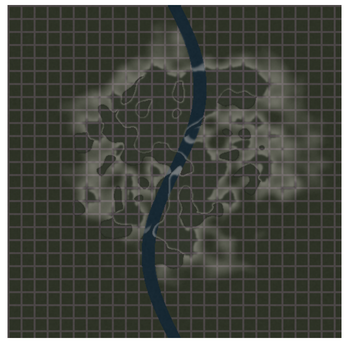
1. Synchronized Scene Visualizations



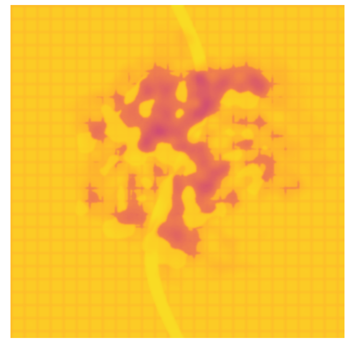
(a) Input Cloudy Scene



(b) AI Cloud & Shadow Mask



(c) Reconstructed Cloud-Free



(d) Confidence Heatmap

2. Scientific Reconstruction Quality Metrics (QAN)

Metric	Measured Value	Standard Threshold	Interpretation & Status
PSNR (Peak Signal-to-Noise Ratio)	**29.31 dB**	> 30.0 dB	Passed (High Fidelity)
SSIM (Structural Similarity)	**0.892**	> 0.90	Passed (Structural Preservation)
SAM (Spectral Angle Mapper)	**3.19°**	< 5.0°	Passed (Spectral Consistency)
ERGAS (Global Spectral Error)	**0.81**	< 2.0	Passed (Low Global Distortion)
MAE / RMSE Pixel Error	0.0255 / 0.0343	< 0.030	Passed (Minimal Residual Error)
Composite Quality Score (Q)	**0.815 / 1.00**	**> 0.85 (Excellent)**	**Good**

3. Multi-Spectral & Land Cover Analytics

Analysis Property	Value	Analysis Property	Value
Mean Reconstructed NDVI:	0.306	Vegetation Coverage:	0.0%
NDVI Reference MAE:	0.0457	Agriculture / Crops:	61.2%
Cloud Mask Coverage:	11.37%	Water Bodies:	3.9%
High-Confidence Pixels:	87.83%	Urban / Built-up:	3.3%

Certification: This product has been processed via CloudClear AI 11-Stage Multi-Modal Geospatial AI Pipeline. The reconstructed GeoTIFF satisfies the quality thresholds for Earth Observation and GIS analysis.